

B10 Series B

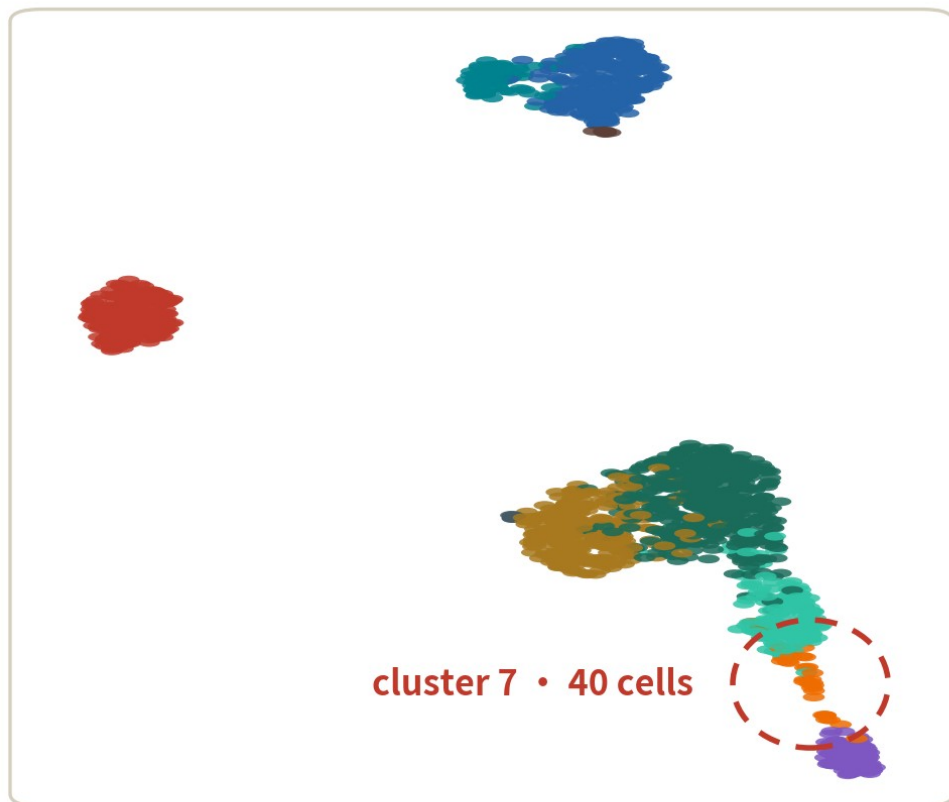
Cell annotation: markers, FindAllMarkers, SingleR / Azimuth

Run manual and automated annotation side by side — and know whom to trust when they disagree.

About 40 min | Prerequisites: B8 (B9 recommended) | Data: PBMC 3k

One cluster, two names

PBMC 3k, ten clusters at res 0.6



simulated data

SingleR says

NK_cell

top similarity: NK 0.42 (T cells 0.40)
delta 0.017 — lowest in the room

markers say

effector CD8 T

CD3E+ CD8A+ (TCR parts NK cells lack)
plus the NKG7 / GNLY cytotoxic program

who is right? — don't pick a side, weigh the evidence first

Cluster 7, 40 cells: SingleR says NK, your markers say effector CD8 T. Who is right?

WHERE THIS EPISODE STARTS

**Annotation is not a lookup —
it is a verdict weighed on evidence.**

Before asking who is right, ask what evidence each side holds. This episode teaches you how to run that trial.

What you will learn

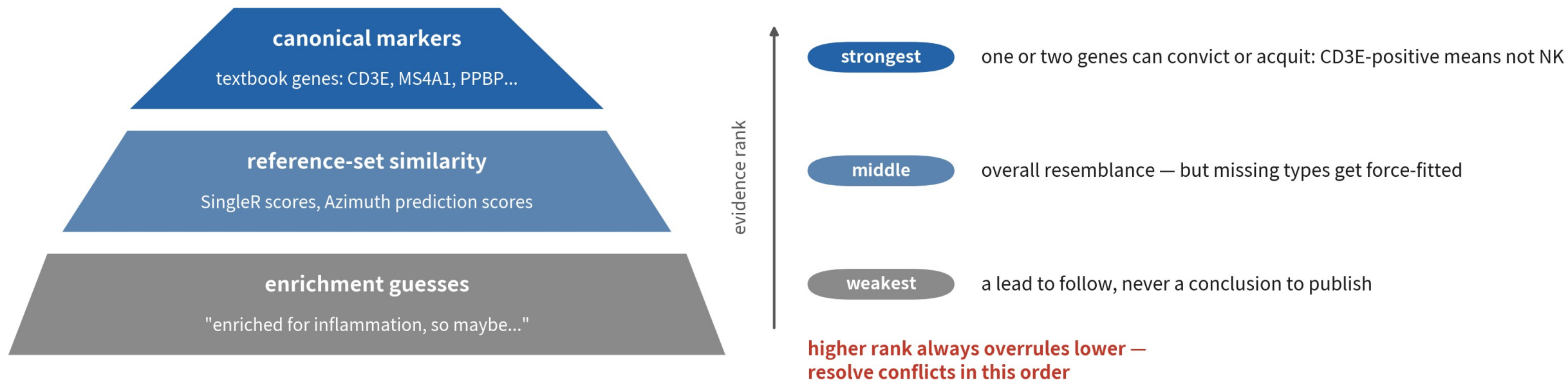
- 1 Run FindAllMarkers and read every column of its output table
- 2 Manually annotate all ten PBMC clusters with a canonical marker panel
- 3 Run SingleR and Azimuth, then adjudicate disagreements with a cross-validation table

01

What annotation really is

Mapping clusters onto known biology — always with uncertainty attached

Three kinds of evidence, three ranks



schematic

- Strongest: canonical markers — textbook-grade signature genes
- Next: reference-set similarity — SingleR / Azimuth scores
- Weakest: enrichment guesses — "this cluster is enriched for..."
- Higher-rank evidence always overrules lower-rank evidence

CORE IDEA

**Every label is a hypothesis.
The only difference is the rank of evidence
behind it.**

"Cluster 3 = B cells" is always an inference, never an observation. Your methods section states the grounds.

02

FindAllMarkers: each cluster's witness list

One call to find the marker genes of all ten clusters

Run FindAllMarkers

```
library(Seurat)
library(dplyr)
set.seed(1234)

pbm.c <- readRDS("output/pbm.c_b08_clustered.rds")
Idents(pbm.c) <- "RNA_snn_res.0.6" # 明確指定，不靠默契

pbm.c.markers <- FindAllMarkers(pbm.c, only.pos = TRUE,
                                min.pct = 0.25,
                                logfc.threshold = 0.25)
dim(pbm.c.markers)
```

```
[1] 3178  7
```

Each cluster is tested one-vs-rest; 3,000+ rows cover all ten clusters

Inspect the output: cluster 3

```
pbm c.m markers |>  
  filter(cluster == 3) |>  
  slice_head(n = 4)
```

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
CD79A	1.1e-182	4.31	0.936	0.041	1.5e-178	3	CD79A
MS4A1	4.4e-180	4.20	0.855	0.053	6.1e-176	3	MS4A1
TCL1A	2.9e-155	4.53	0.622	0.023	3.9e-151	3	TCL1A
CD79B	6.2e-149	3.87	0.891	0.142	8.5e-145	3	CD79B

CD79A and MS4A1 — parts of the B-cell receptor. This cluster wears its identity on its face

The table, column by column

top three rows of cluster 3 (B-cell markers)

read first: fold change (log2)

4.31 = about 20-fold

read second: fraction in vs out

0.94 vs 0.04 — the wider the better

	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
CD79A	1.1e-182	4.31	0.94	0.04	1.5e-178	3	CD79A
MS4A1	4.4e-180	4.20	0.86	0.05	6.1e-176	3	MS4A1
TCL1A	2.9e-155	4.53	0.62	0.02	3.9e-151	3	TCL1A

raw p-value: never use directly
housekeeping genes ace it too (trap 1)

after Bonferroni: the final gate
multiplied by all 13,714 genes

which gene of which cluster
the gene column is authoritative

Read avg_log2FC and the pct gap first, p_val_adj last — the traps section explains why

Top five per cluster, one roll call

```
top5 <- pbmc.markers %>%  
  group_by(cluster) %>%  
  slice_max(avg_log2FC, n = 5) %>%  
  ungroup()  
  
top5 %>% filter(cluster == 9) %>%  
  select(gene, avg_log2FC, pct.1, pct.2)
```

	gene	avg_log2FC	pct.1	pct.2
1	PPBP	8.58	1.000	0.011
2	PF4	7.24	1.000	0.010
3	GNG11	6.94	0.929	0.019
4	SDPR	6.65	0.929	0.011
5	TUBB1	6.26	0.857	0.005

Cluster 9: PPBP and PF4 — textbook platelet markers, pct.2 near zero

What this section did

One witness list



3,178 rows holding
every cluster's
candidate markers

Three columns to read



avg_log2FC, pct.1 vs
pct.2, p_val_adj — in
that order

The table cannot speak



the list must face
known biology —
visualization first

03

The marker-visualization trio

FeaturePlot, VlnPlot, DotPlot — three ways to ask one question

The trio, one line each

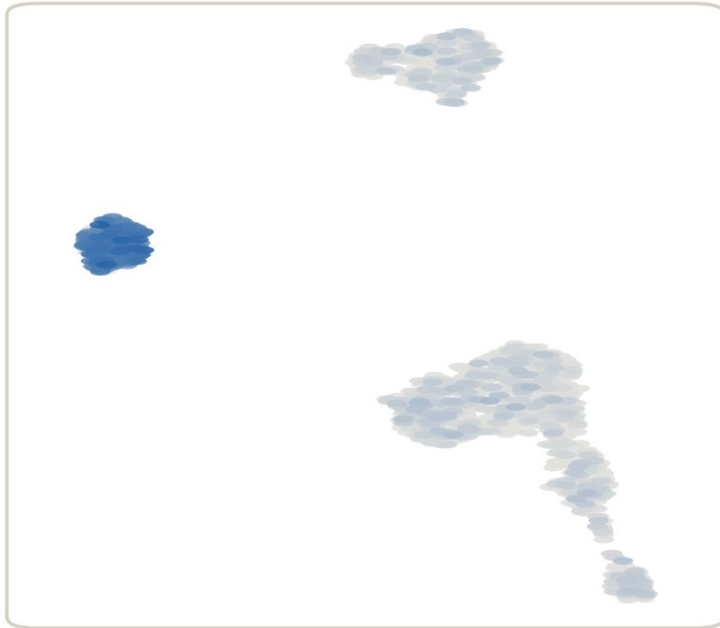
```
# 在哪裡亮：表現量塗在 UMAP 上  
FeaturePlot(pbm c, features = "M S4A1")  
  
# 各群分布：一群一把小提琴  
VlnPlot(pbm c, features = "M S4A1", pt.size = 0)  
  
# 多基因 × 多群總覽：大小 = 比例、深淺 = 強度  
DotPlot(pbm c, features = c("M S4A1", "CD 79A", "CD 3E", "NKG 7")) +  
  RotatedAxis()
```

A5 taught you to read these plots; today you draw them

One marker, three views

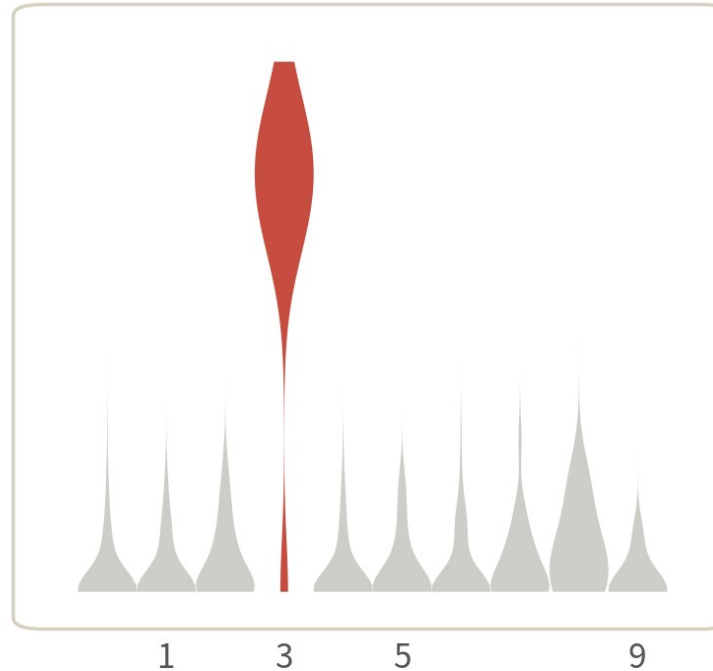
simulated data • marker: MS4A1 (B cells)

FeaturePlot: where it lights



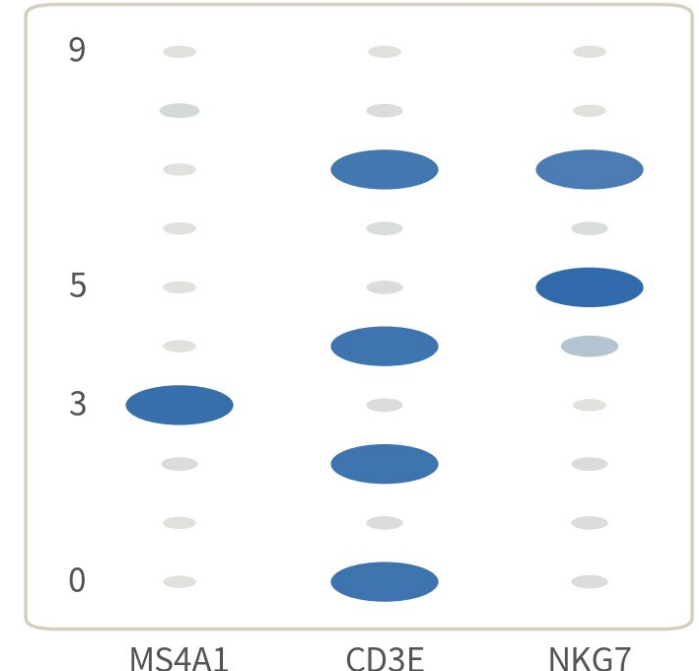
only the B cluster (3) lights up

VlnPlot: per-cluster distribution



the whole cluster is high — not a few loud cells

DotPlot: multi-gene overview



size = fraction, colour = level

Location, distribution, overview — each covers the others' blind spots; any one alone can mislead

Which plot when

Tool	What it gives you	When to use it
FeaturePlot	expression painted on the embedding — where it lights up	check that the bright region matches the cluster
VlnPlot	one violin per cluster — distribution and bimodality	tell a truly high cluster from a few loud cells
DotPlot	genes × clusters; size = fraction, colour = level	one-page panel overview; a staple of paper figures

04

Manual annotation: walk the panel cluster by cluster

The canonical PBMC panel versus ten clusters, one verdict at a time

The canonical PBMC marker panel

type (cluster)	canonical markers	one-line mnemonic
 Naive CD4 T (0)	CD3E • IL7R • CCR7	T backbone + naive-state CCR7
 Memory CD4 T (2)	CD3E • IL7R • S100A4	same backbone, S100A4 replaces CCR7
 CD8 T (4)	CD3E • CD8A	T backbone + CD8A
 Effector CD8 T (7)	CD3E • CD8A • NKG7 • GNLY	T backbone + cytotoxic set — mimics NK
 NK (5)	NKG7 • GNLY (CD3E-negative)	cytotoxic set without a TCR
 B (3)	MS4A1 • CD79A	parts of the B-cell receptor
 CD14 monocyte (1)	CD14 • LYZ	classical; LYZ blazing high
 FCGR3A monocyte (6)	FCGR3A • MS4A7	non-classical (CD16)
 Dendritic cell (8)	FCER1A • CST3	professional antigen presenter
 Platelet (9)	PPBP • PF4	strictly not a cell — no nucleus

Worth memorizing — anyone working with PBMC will use this table forever

Deploy the panel

```
panel<- c("CD3E", "IL7R", "CCR7", "S100A4", # T 與 CD4 亞群
          "CD8A",          # CD8 T
          "MS4A1",         # B
          "CD14", "LYZ",    # 古典單核球
          "FCGR3A", "MS4A7", # 非古典單核球
          "NKG7", "GNLY",   # NK 與胞殺程式
          "FCER1A", "CST3",  # DC
          "PPBP")           # 血小板
```

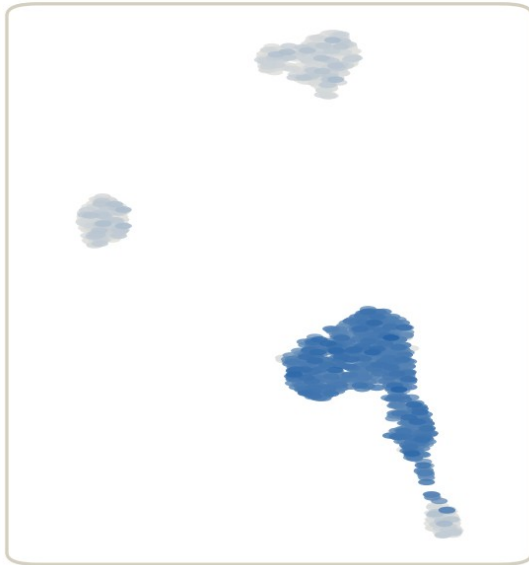
```
FeaturePlot(pbm c, features = c("CD3E", "MS4A1", "CD14", "NKG7"))
DotPlot(pbm c, features = panel) + RotatedAxis()
```

CD4 mRNA drops out easily; use IL7R to support CD4 T calls — protein and mRNA differ

The walk, cluster by cluster

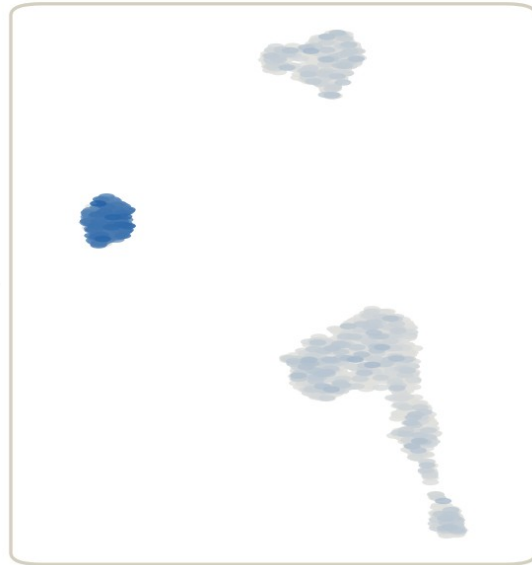
simulated data

CD3E



whole lower-left lights
→ T lineage (0/2/4/7)

MS4A1



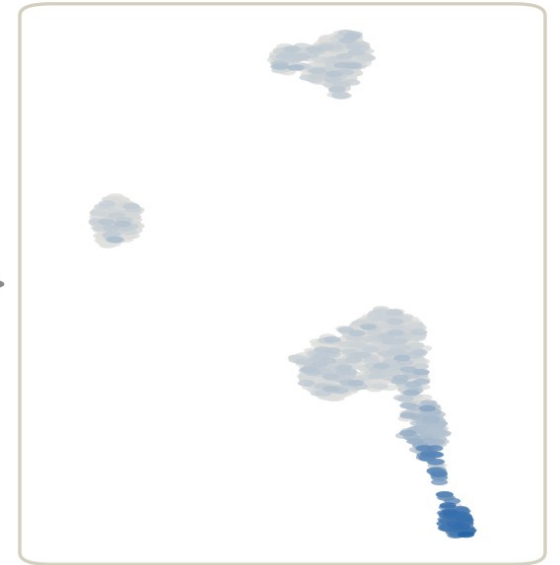
one block only
→ B (3)

CD14



one upper lobe
→ classical mono (1)

NKG7



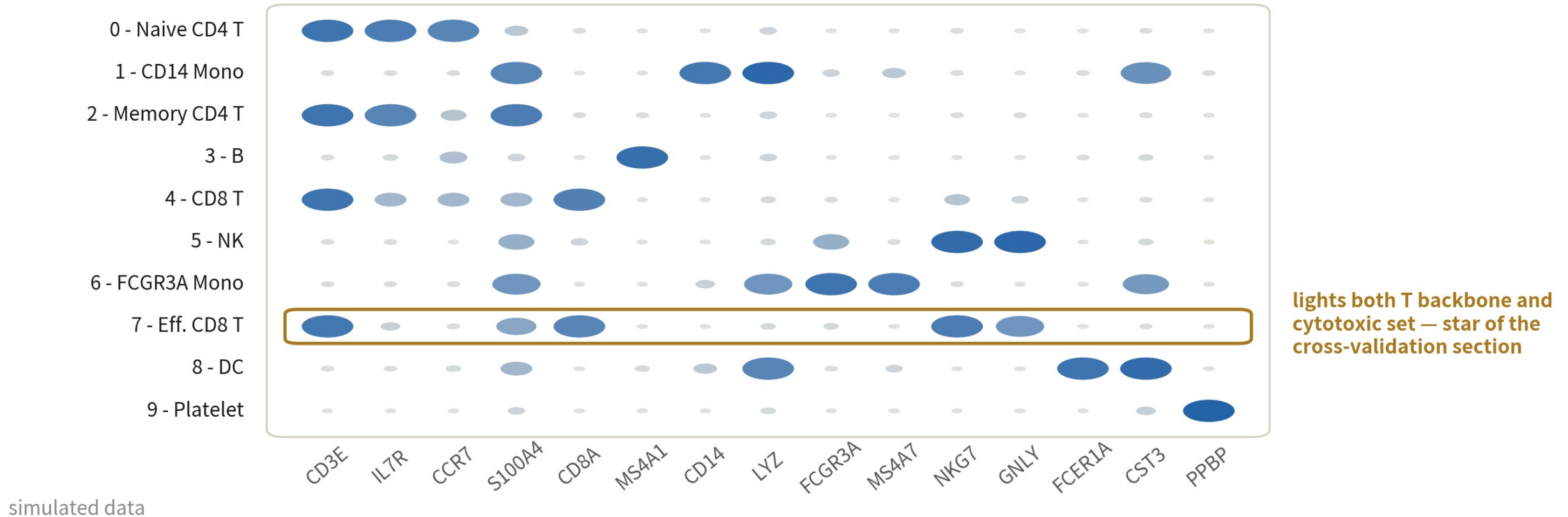
NK lights — plus cluster 7
→ 7 needs a second look

one marker, one question: where does it light, who does it map to, any counter-evidence?

Ask each cluster: which markers light up, who do they map to, and is there counter-evidence?

DotPlot panorama: ten clusters at once

dot size = fraction of cells expressing | colour = mean expression



Bright blocks along the diagonal mean every cluster has evidence; cluster 7 lights both T and cytotoxic sets — remember it

Hang the names

```
new.ids <- c("Naive CD4 T", "CD14 Mono", "Memory CD4 T",  
            "B", "CD8 T", "NK", "FCGR3A Mono",  
            "Effector CD8 T", "DC", "Platelet")  
names(new.ids) <- levels(pbm.c)  
pbm.c <- RenameIds(pbm.c, new.ids)  
pbm.c$celltype.manual <- Idents(pbm.c) # 存進 metadata  
table(pbm.c$celltype.manual)
```

Naive CD4 T	CD14 Mono	Memory CD4 T	B
711	436	467	344
CD8 T	NK	FCGR3A Mono	Effector CD8 T
271	165	158	40
DC	Platelet		
32	14		

Ten clusters, nine cell types — stored as a metadata column, cluster IDs untouched

05

Automated annotation: SingleR and Azimuth

Let reference datasets vote — then learn to read their confidence

SingleR: similarity to a reference

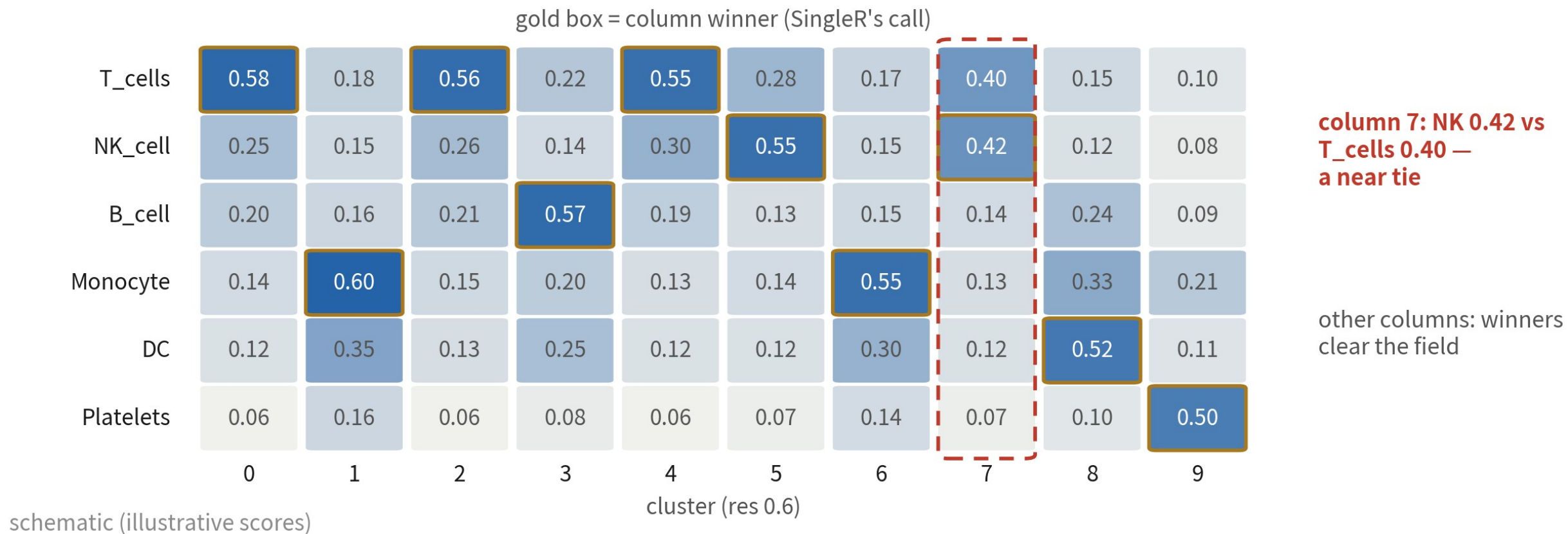
```
# BiocManager::install(c("SingleR", "celldex"))
library(SingleR)
library(celldex)
ref <- HumanPrimaryCellAtlasData() # HPCA : 37 種主標籤

pred <- SingleR(test = GetAssayData(pbm c, layer = "data"),
               ref = ref, labels = ref$labelmain,
               clusters = pbm c$RNA_snn_res.0.6)
pred$labels
```

```
[1] "T_cells" "Monocyte" "T_cells" "B_cell"
[5] "T_cells" "NK_cell" "Monocyte" "NK_cell"
[9] "DC"      "Platelets"
```

clusters = per-cluster mode; the 8th label (cluster 7) reads NK_cell — the cold open's origin

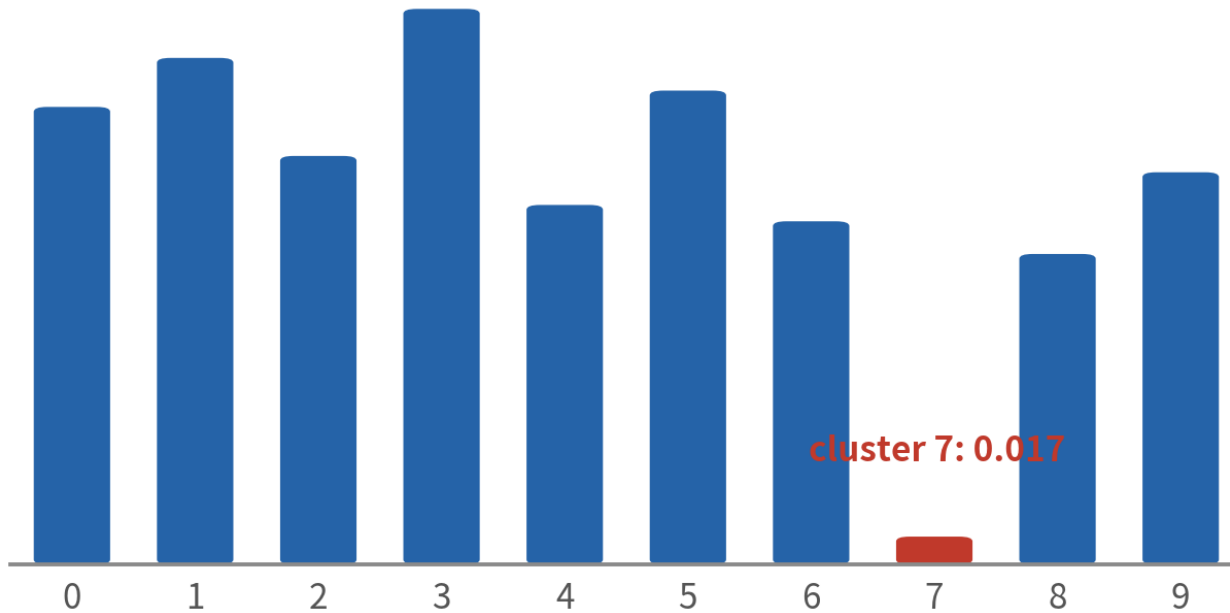
The score heatmap: reading the scores



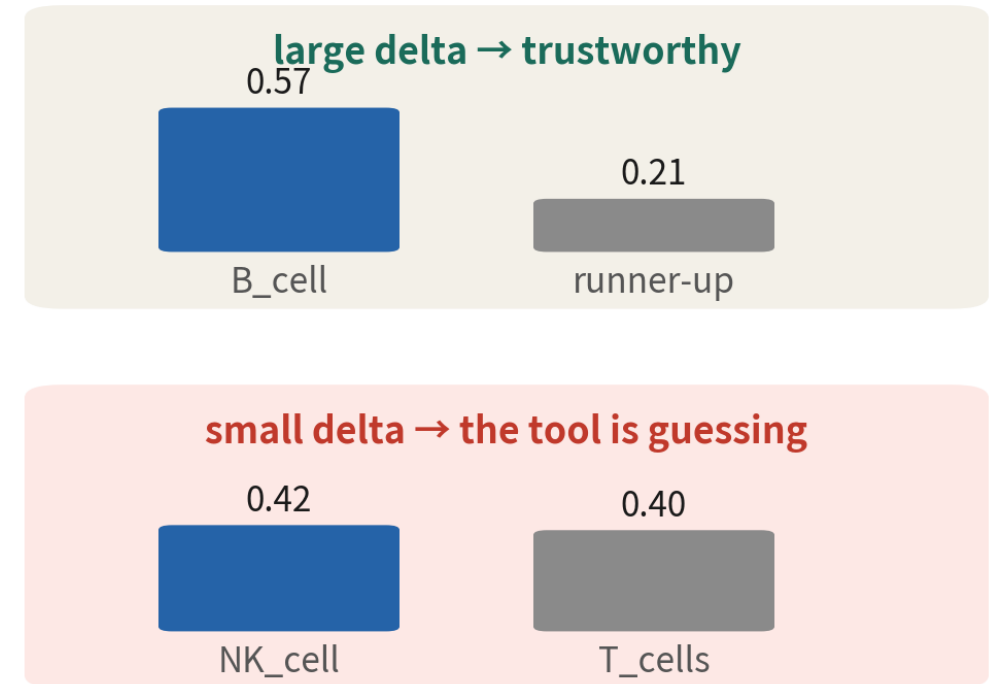
The question is not who scores highest, but by how much the winner clears the runner-up

Delta: SingleR's confidence

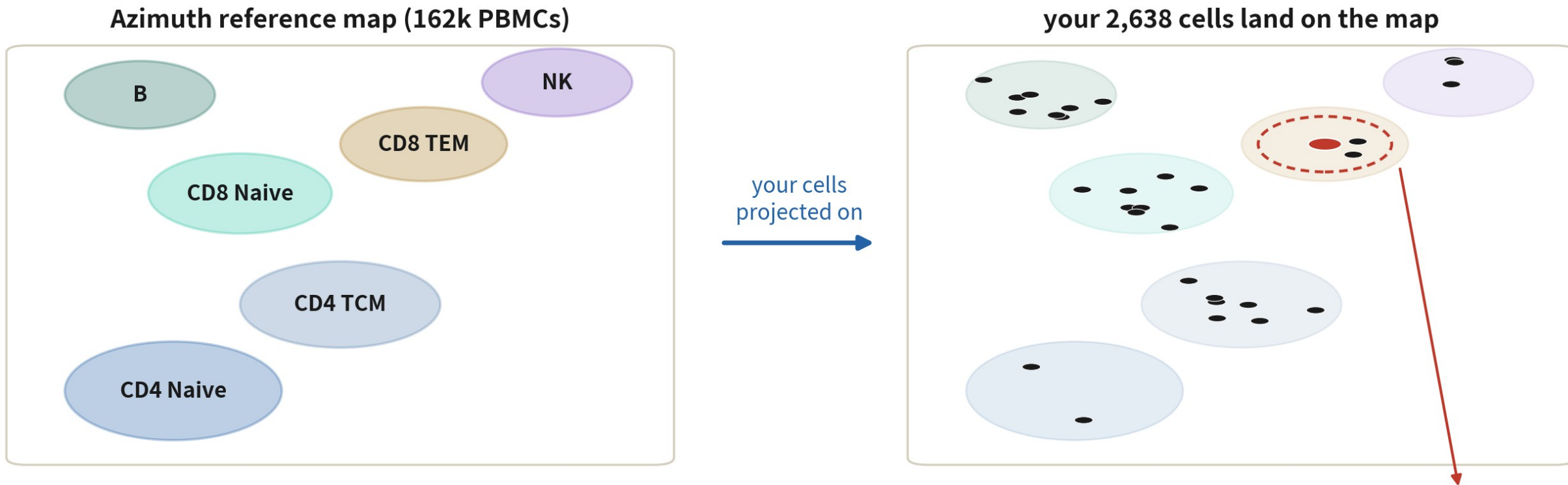
delta per cluster (top score minus runner-up)



Cluster 7's delta is 0.017, the lowest in the room — SingleR itself admits it is unsure



Azimuth: placing cells on a reference map



schematic - no re-clustering, just localization: GPS dots on an existing map

nearby reference cells vote → label + prediction score

Not re-clustering — your cells are projected onto a curated map and read off their location

Run Azimuth

```
# remotes::install_github("satijalab/azimuth")
library(Azimuth)
pbmc <- RunAzimuth(pbmc, reference = "pbmc_ref")

head(pbmc@meta.data[, c("predicted.celltype.l2",
                        "predicted.celltype.l2.score")], 3)
```

	predicted.celltype.l2	predicted.celltype.l2.score
AAACATACAACCAC -1	CD4 Naïve	0.943
AAACATTGAGCTAC -1	B naïve	1.000
AAACATTGATCAGC -1	CD4 TCM	0.741

l1 / l2 / l3 labels go coarse to fine; the score measures agreement with the reference, not correctness

SingleR vs Azimuth

Not either-or — they fail differently, which is why you run both

SingleR

Similarity to any reference

- accepts bulk and microarray references — very flexible
- outputs scores and deltas, per cluster or per cell
- only as good as the reference you pick

Azimuth

Projection onto a curated atlas

- high-quality official references for common tissues
- three label levels, down to subsets like CD8 TEM
- no map for your tissue means no Azimuth

06

Cross-validation: three parties, one table

Manual, SingleR and Azimuth side by side — verdicts cluster by cluster

The three-way cross-validation table

cluster (n)	manual	SingleR	Azimuth (l2 majority)	verdict
0 (711)	Naive CD4 T	T_cells	CD4 Naive	✓ agree
1 (436)	CD14 Mono	Monocyte	CD14 Mono	✓ agree
2 (467)	Memory CD4 T	T_cells	CD4 TCM	✓ agree
3 (344)	B	B_cell	B naive	✓ agree
4 (271)	CD8 T	T_cells	CD8 Naive	✓ agree
5 (165)	NK	NK_cell	NK	✓ agree
6 (158)	FCGR3A Mono	Monocyte	CD16 Mono	✓ agree
7 (40)	Effector CD8 T	NK_cell	CD8 TEM	⚠ to trial
8 (32)	DC	DC	cDC2	✓ agree
9 (14)	Platelet	Platelets	Platelet	✓ agree

"T_cells vs CD4 Naive" is a granularity gap, not a conflict; the only real fight is cluster 7

Nine clusters agree three ways; the only fight is cluster 7 — the cold open cluster

The trial: cluster 7 versus real NK

```
# SingleR 說 NK、Azimuth 說 CD8 TEM、手動說效應 CD8 T。  
# 讓 7 號跟公認的 NK (5 號) 當面對質：  
Idents(pbm c) <- "RNA_snn_res.0.6"  
vs.nk <- FindMarkers(pbm c, ident.1 = 7, ident.2 = 5,  
                      min.pct = 0.25)  
vs.nk[c("CD3E", "CD8A", "GNLY", "NKG7"),  
       c("avg_log2FC", "pct.1", "pct.2")]
```

	avg_log2FC	pct.1	pct.2
CD3E	3.94	0.975	0.061
CD8A	3.11	0.850	0.024
GNLY	-1.22	0.700	0.964
NKG7	-0.48	0.950	0.994

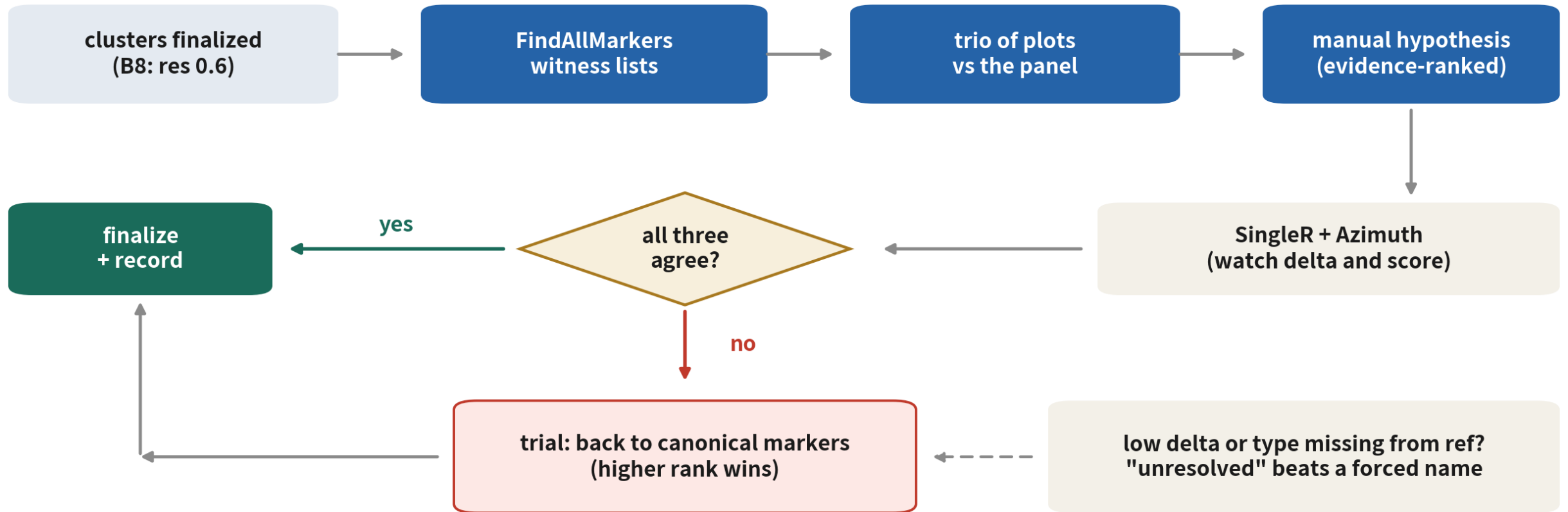
CD3E/CD8A means T cell, full stop; a shared cytotoxic program fools similarity scores but not canonical markers

FOR YOUR METHODS SECTION

**"Clusters were annotated manually with canonical markers,
cross-validated by SingleR and Azimuth;
disagreements were resolved by marker evidence."**

Attach the cross-validation table as supplementary material and reviewers have nothing left to ask.

The whole annotation workflow, one picture



Hypothesize manually, verify automatically, adjudicate by evidence rank — this works for any tissue

07

Where people get hurt

Three traps — each demonstrated for real

Trap 1: the smallest p-value is the best marker

What it is

Sorting the FindAllMarkers table by p-value and taking the top rows as markers.



Why it happens

Statistics class said smaller p means more significant, so sorting by p feels natural.

What it costs you

With 2,000+ cells, housekeeping genes expressed everywhere also get astronomically small p-values; your "best marker" is one that everyone expresses and distinguishes nothing.

Two genes, equally dazzling p-values

two candidate markers for cluster 5 (NK) — both p-values on the floor

B2M



p_val_adj = 3.1e-146

pct.1  1.00

pct.2  0.99

avg_log2FC = 0.58

housekeeping: everyone expresses it

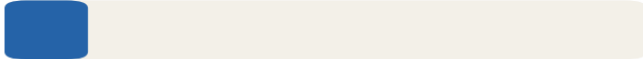
p is
a tie

NKG7



p_val_adj = 1.9e-151

pct.1  0.99

pct.2  0.13

avg_log2FC = 4.05

true marker: only NK is high

sort by p and you pick B2M; sort by FC and pct gap and you pick NKG7

The p-values cannot tell them apart; the pct gap and log2FC can, instantly. Read FC, then pct gap, then p

Trap 2: taking automated output as the conclusion

What it is

Using whatever label SingleR or Azimuth returns, with no further checks.



Why it happens

The tools are famous and the scores look high — surely more reliable than your own judgment.

What it costs you

Cell types absent from the reference get force-fitted to the nearest label — these tools have no "none of the above", so your platelets quietly become monocytes.

Live demo: platelets meet a reference that has never seen one

```
# DICE 參考集：只有 T / B / NK / 單核球——沒有 DC、沒有血小板
ref.dice <- DatabaseImmuneCellExpressionData()
pred.d <- SingleR(test = GetAssayData(pbm.c, layer = "data"),
  ref = ref.dice, labels = ref.dice$label.main,
  clusters = pbm.c$RNA_snn_res.0.6)
# 8 號 (DC) 與 9 號 (血小板) 拿到什麼標籤?
pred.d$labels[9:10]
round(pred.d$delta.next[9:10], 3)
```

```
[1] "Monocytes" "Monocytes"
[1] 0.021 0.008
```

Both clusters get stuffed into 'Monocytes' with deltas on the floor — a low delta is your only alarm

Trap 3: coarse markers at subcluster resolution

What it is

After splitting T cells into four subclusters, still using pan-markers like CD3E to tell them apart.



Why it happens

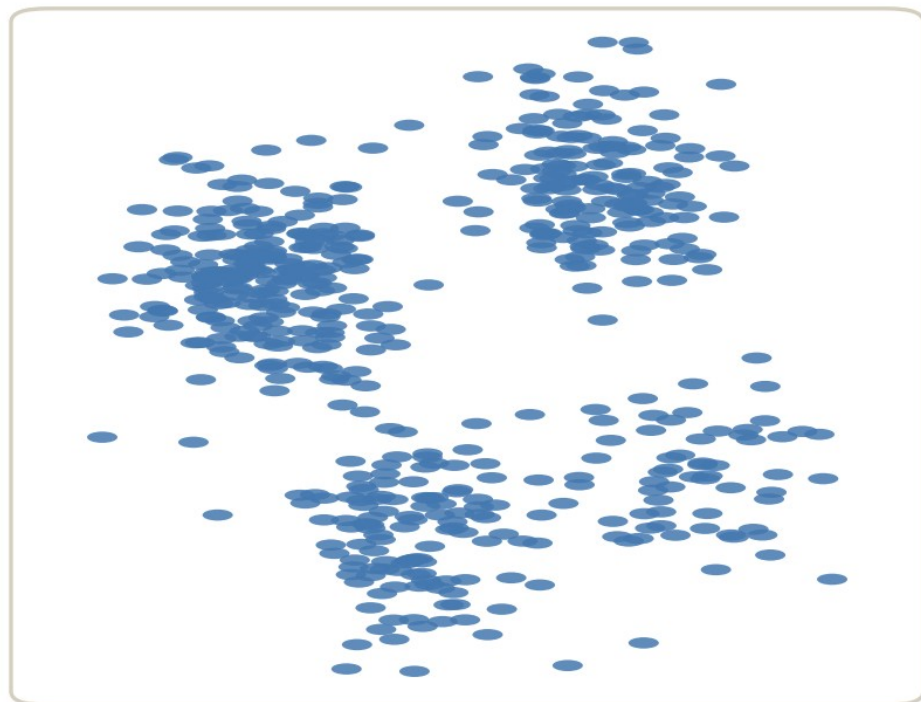
CD3E is the most famous T-cell marker — famous enough to make you forget its resolution.

What it costs you

All four subclusters are CD3E-positive and all get labelled "T cell" — the split was pointless. Subcluster questions need subcluster panels.

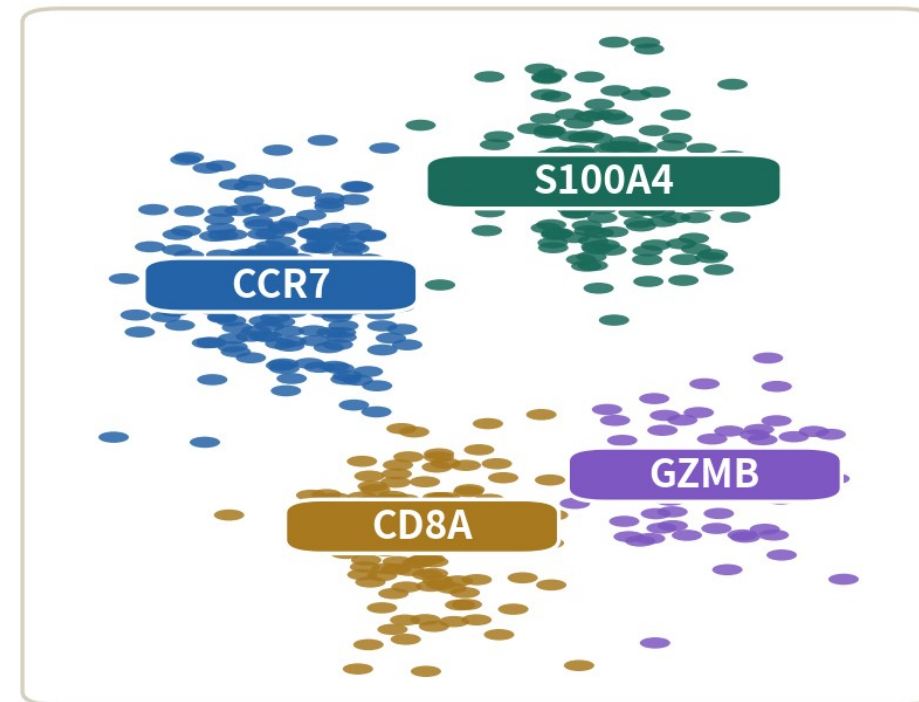
Marker resolution must match cluster resolution

pan-marker: CD3E



all four light up → all labelled "T cell"
the split was pointless
simulated data

subset markers: CCR7 / S100A4 / CD8A / GZMB



one marker per subset → names that stick

swap panels
with resolution



Pan-markers separate lineages; subset markers separate subsets — swap panels as resolution deepens

TRAPS, SUMMED UP

**Tools hand you scores.
The name is always yours to give.**

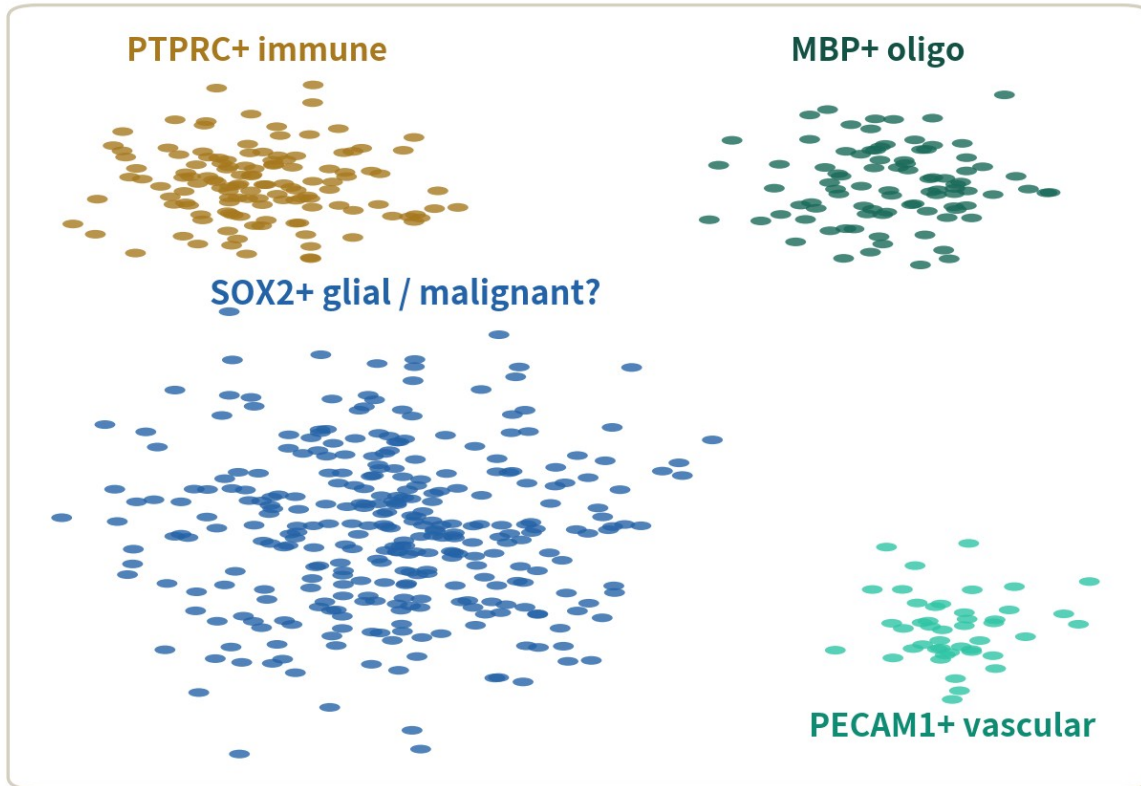
P-values, similarities and prediction scores are only evidence — you sign the verdict, and you own the errors.

05+

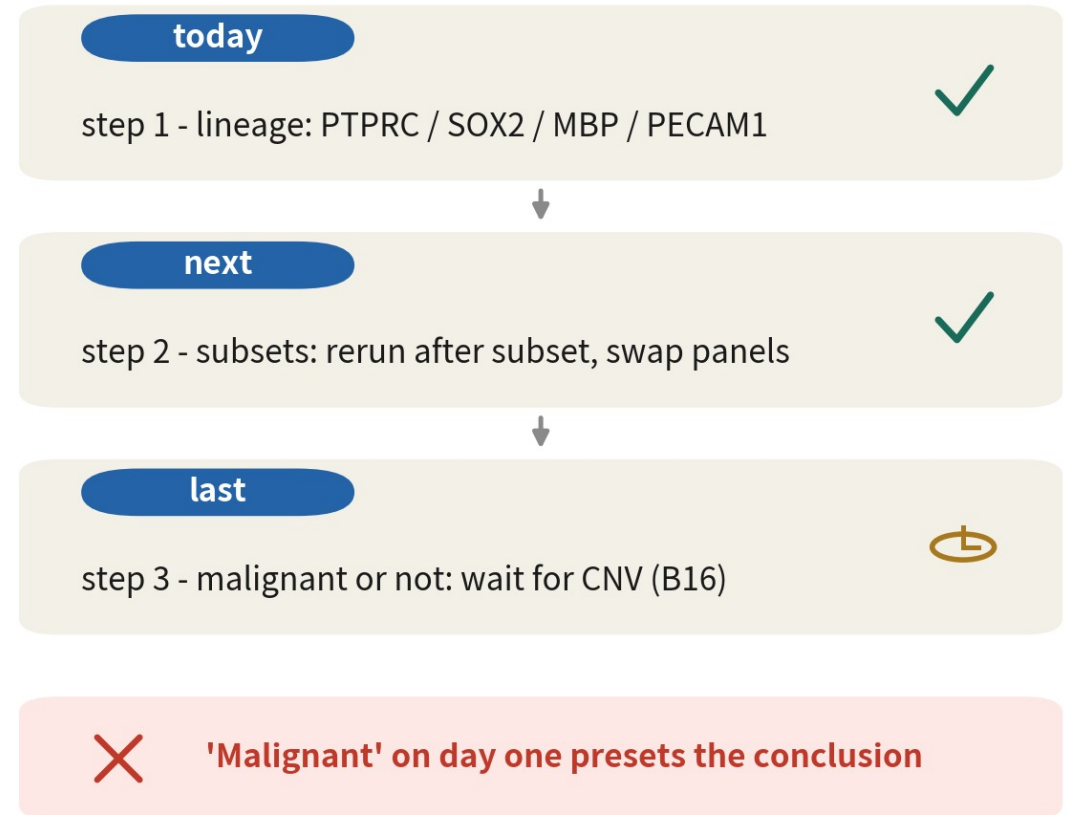
Strategy Room: Complex Samples

The same annotation on a tumour — order, panels, references and names all need a rethink

Order of annotation: lineage first, malignancy later



tumour UMAP: four lineage continents — the glial one cannot be called malignant today



schematic

Markers settle lineage only; malignancy waits for B16's CNV evidence — 'Malignant' on day one presets the conclusion

Hierarchical annotation: subsets rerun the pipeline

Subset the immune compartment, recompute HVG, PCA, clustering and UMAP — then switch panels

Why rerun



global HVGs and top PCs are consumed by malignant-vs-immune; subset signal never makes the list

Panels move down a level



P2RY12/TMEM119 call microglia, CD163/LYZ blood-derived TAMs; CD8A, FOXP3, PD1 split the T cells

Two hard lines



don't subdivide below ~200 cells; malignant 'subsets' are scored as states, not clustered

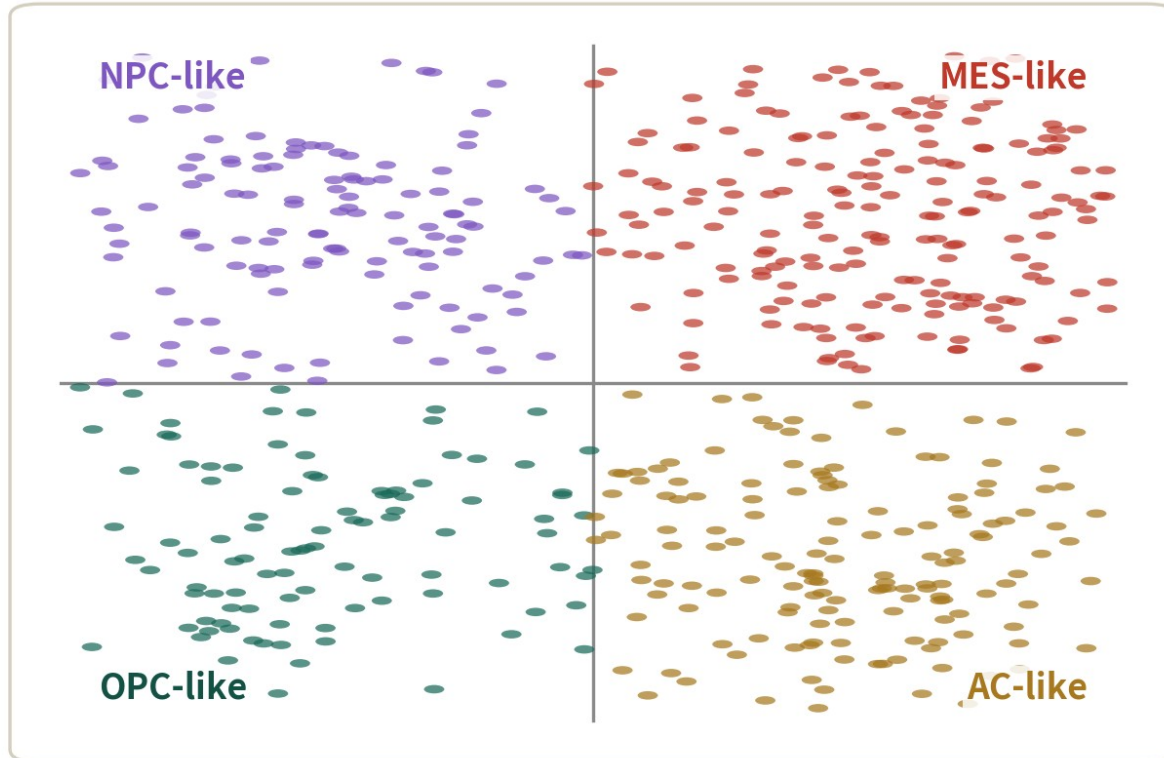
Picking references: different tumour, different atlas

Choose by who you are annotating — one tumour dataset, three separate decisions

Annotating whom	Which reference	The trap
TME immune cells	immune-dedicated references: Monaco, CellTypist immune atlases	broad sets like HPCA lack subset resolution
Normal stroma & lineages	HPCA or an organ atlas; oligo and endothelium rarely disputed	gaps still get force-fitted — watch the delta
Malignant cells	no reference contains your tumour	always force-fitted to the nearest normal type (GBM→astrocyte) — trap 2, magnified

Malignant cells get states, not types

inside the malignant continent: states are continuous, cells blur across quadrants



← OPC/NPC-like AC/MES-like →

AddModuleScore

gene-set mean minus control mean - fast, depth-sensitive

UCell

rank-based - most robust to depth; trust it in conflicts

AUCell

AUC over the top-5% ranks - the SCENIC engine



**scores describe state, never prove malignancy -
the states come from confirmed-malignant cells**

simulated data

Neftel's four states are gene-set scores: continuous and comparable across patients; scores describe state, never prove malignancy

THE NAMING RULE

**Write names only as far as evidence goes:
lineage + "possibly malignant".**

Store three metadata layers — l1 compartment, l2 type, l3 state; the malignancy call waits for B16's CNV evidence.

Episode checklist



marker table read order: FC, pct gap, p_val_adj



every cluster backed by canonical markers



SingleR run; score heatmap and deltas inspected



Azimuth run; prediction scores inspected



cross-validation table built; conflicts adjudicated



final labels stored in metadata; original cluster IDs kept

All six boxes ticked before your annotation is final

ONE-LINE SUMMARY

**Hypothesize manually, verify automatically —
when they fight, evidence rank decides.**

Canonical markers > reference similarity > enrichment guesses. The whole episode is this one ordering.

Check yourself 1

In the FindAllMarkers output, what are pct.1 and pct.2?

- A Mean expression in this cluster and in the others
- B The fraction of cells expressing the gene, in this cluster vs the rest
- C Loadings on PC1 and PC2
- D Raw and adjusted p-values

Answer B. pct.1 is the fraction of cells expressing the gene within the cluster; pct.2 is the fraction across all other clusters combined. A good marker shows high pct.1 and low pct.2 — nearly everyone in my cluster, almost no one elsewhere.

Check yourself 2

Which set of numbers looks most like a good marker?

- A $p=1e-250$, $\log_2FC=0.3$, $pct.1=0.99$, $pct.2=0.97$
- B $p=1e-90$, $\log_2FC=3.8$, $pct.1=0.94$, $pct.2=0.05$
- C $p=0.04$, $\log_2FC=1.2$, $pct.1=0.15$, $pct.2=0.02$
- D $p=1e-120$, $\log_2FC=-2.5$, $pct.1=0.10$, $pct.2=0.88$

Answer B. Big fold change, expressed by nearly all cells inside, almost none outside — all three conditions met; the p-value merely needs to pass. A is a housekeeping profile (trap 1); C is expressed by only 15% of cells; D is an inverse marker, high everywhere else.

Check yourself 3

SingleR calls a cluster NK, but it is clearly CD3E- and CD8A-positive. The right move is?

- A Trust SingleR — it looks at more genes
- B Trust the markers: canonical evidence outranks similarity — call it effector CD8 T and record the grounds
- C Report both labels in the paper
- D Delete the cluster to avoid controversy

Answer B. CD3E is part of the TCR complex, which NK cells lack — textbook-grade exclusionary evidence. Effector CD8 T and NK share the cytotoxic program (NKG7, GNLY), which fools similarity scores but not canonical markers. Check the delta too: such clusters sit at the bottom of the distribution.

NEXT EPISODE

B11: Multi-sample integration and batch diagnostics

The single-sample pipeline is complete. But your experiment has treatment, control, three batches — two samples sit as separate islands on the UMAP: batch effect to remove, or biology to keep? Next: integration, and when not to integrate.